MySQL Database Connection in Python

1. Definition

MySQL database connection in Python is the process of connecting a Python program to a MySQL database to perform operations such as inserting, retrieving, updating, and deleting data.

2. Use Cases

- Student management systems
- Backend development
- Data storage and retrieval
- Data analysis workflows

Example: A school stores student details in a MySQL database and Python is used to manage and process that data.

3. Installation

```
pip install mysql-connector-python
```

4. Basic Syntax

```
import mysql.connector

conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="your_password",
    port=3306,
    database="your_db_name"
)

cursor = conn.cursor()
```

5. Example: Student Database

Step 1: Connect to MySQL

```
import mysql.connector

conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="root123",
    port=3306
)
cursor = conn.cursor()
```

Step 2: Display Databases

```
cursor.execute("SHOW DATABASES")

for db in cursor:
    print(db)
```

Step 3: Create Database

```
cursor.execute("CREATE DATABASE student_db")
```

Step 4: Use Database

```
conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="root123",
    port=3306,
    database="student_db"
)

cursor = conn.cursor()
```

Step 5: Create Table

```
cursor.execute(
"CREATE TABLE student (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(50),
    marks INT,
    city VARCHAR(50)
)"
)
```

Step 6: Insert One Student

```
query = "INSERT INTO student (name, marks, city) VALUES (%s, %s, %s)"
values = ("Raki", 85, "Bangalore")

cursor.execute(query, values)
conn.commit()
```

Step 7: Insert Multiple Students

```
query = "INSERT INTO student (name, marks, city) VALUES (%s, %s, %s)"

values = [
    ("Anil", 78, "Mysore"),
    ("Priya", 92, "Chennai"),
    ("Rahul", 67, "Delhi")
```

```
]

cursor.executemany(query, values)
conn.commit()
```

Step 8: Select Data

```
cursor.execute("SELECT * FROM student")

result = cursor.fetchall()

for row in result:
    print(row)
```

Step 9: Update Data

```
query = "UPDATE student SET marks = %s WHERE name = %s"
values = (95, "Raki")

cursor.execute(query, values)
conn.commit()
```

Step 10: Delete Data

```
query = "DELETE FROM student WHERE name = %s"
values = ("Rahul",)

cursor.execute(query, values)
conn.commit()
```

6. Important Notes

- Use conn.commit() after INSERT, UPDATE, and DELETE operations
- Use %s as placeholder for values in queries
- Default MySQL port is 3306
- Always close cursor and connection after completing operations

```
cursor.close()
conn.close()
```